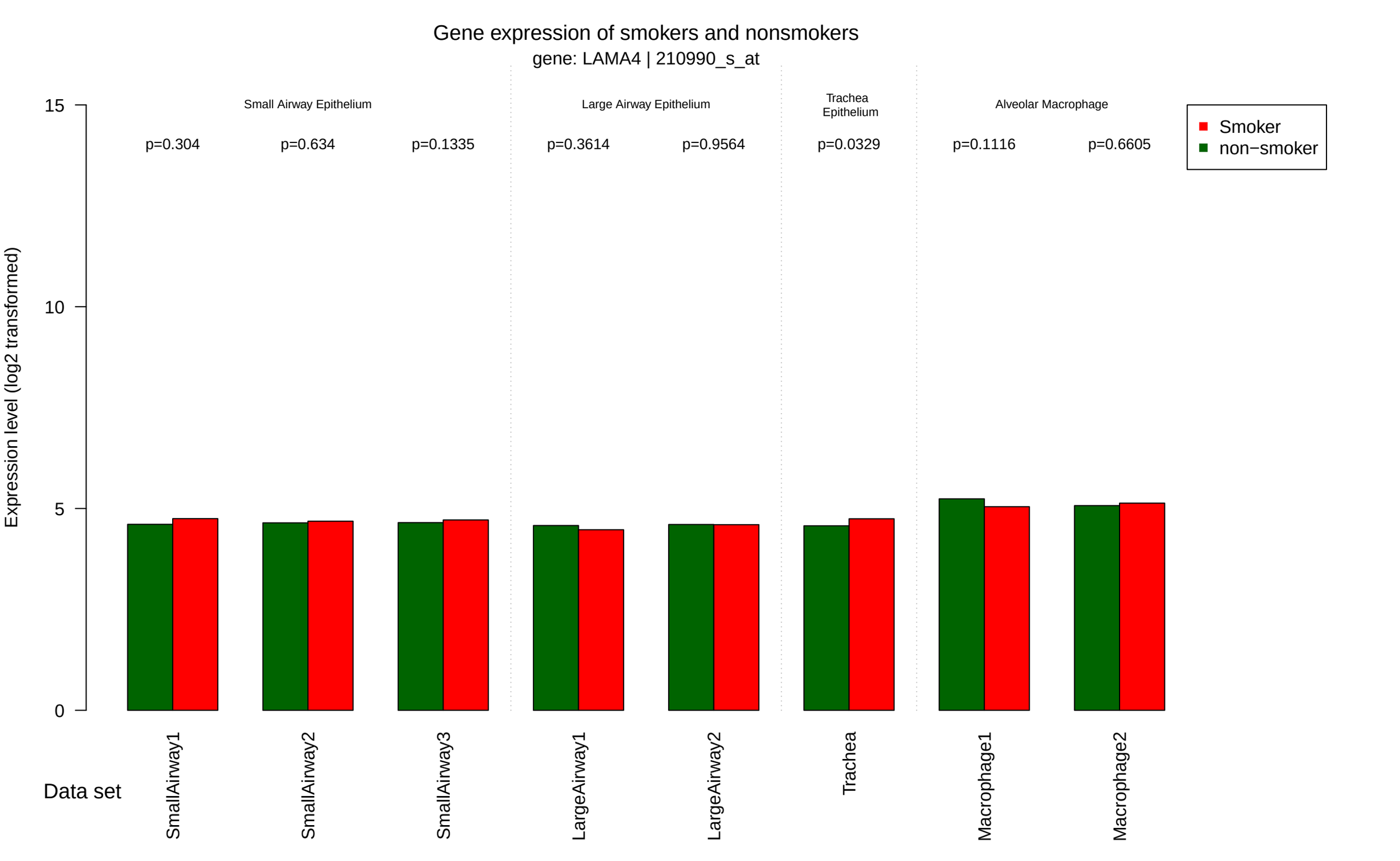
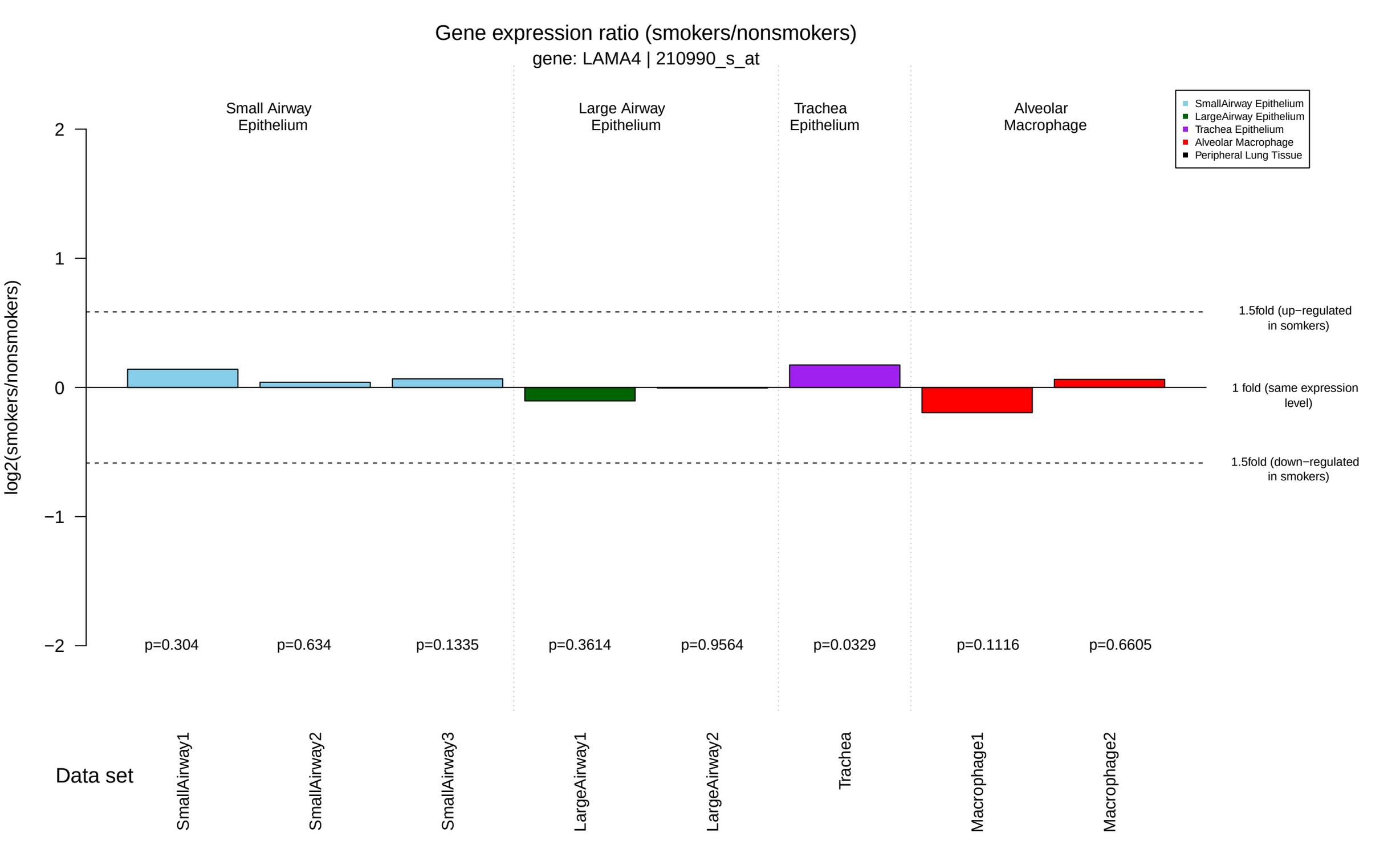
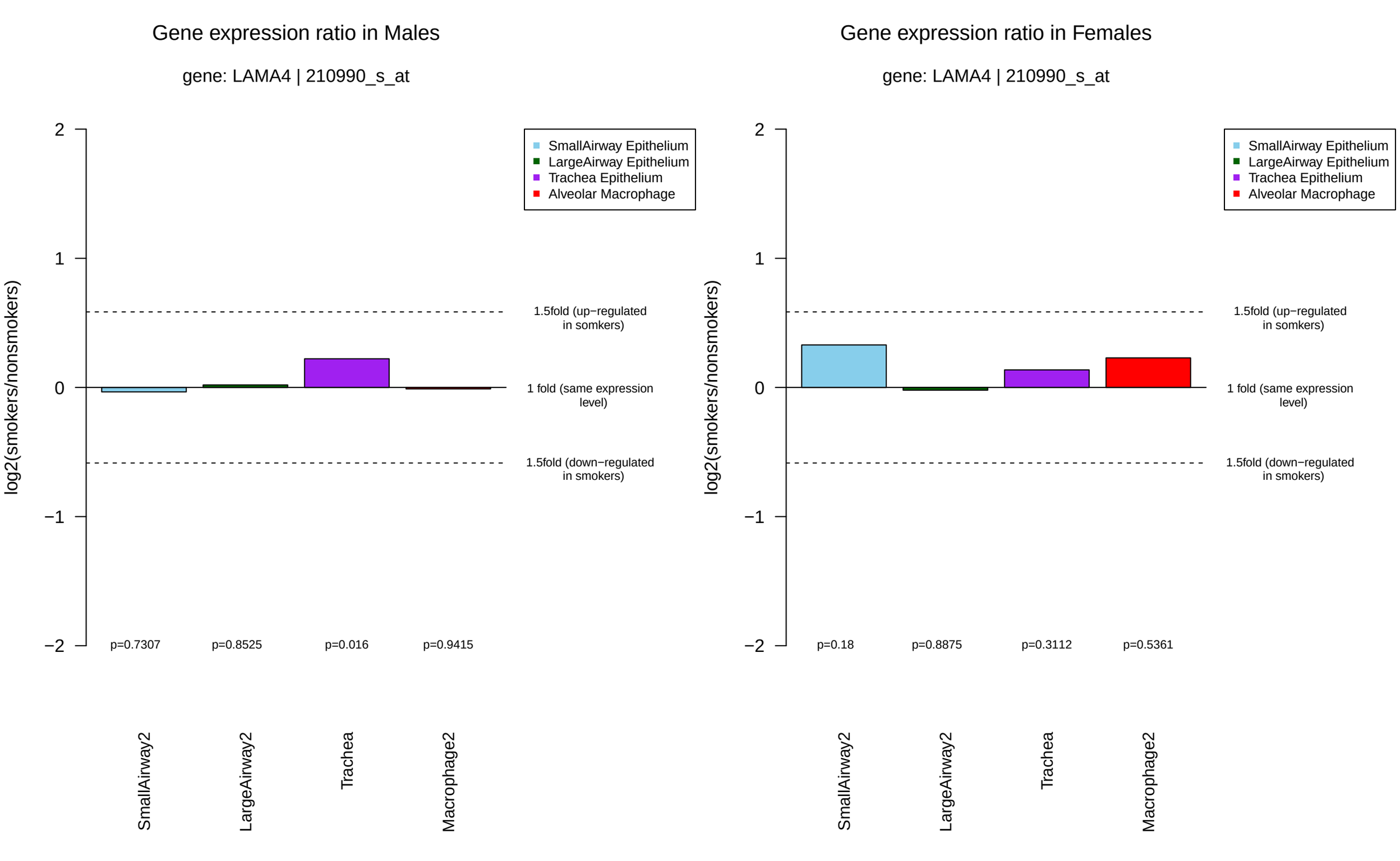
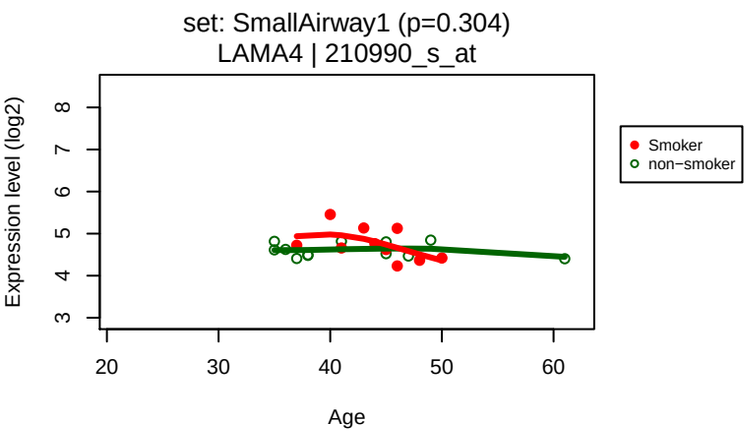
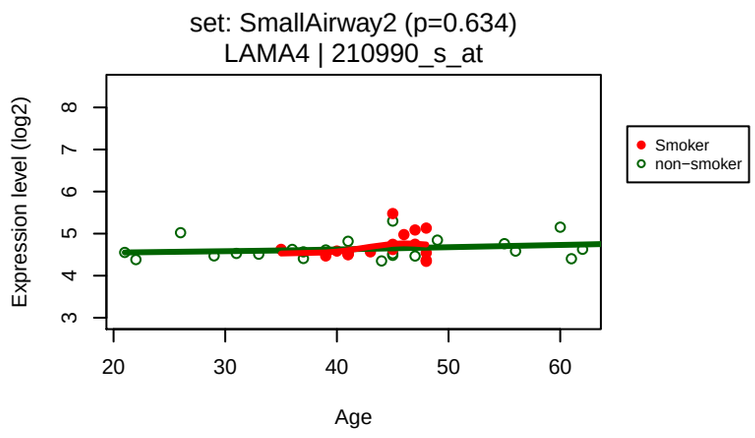
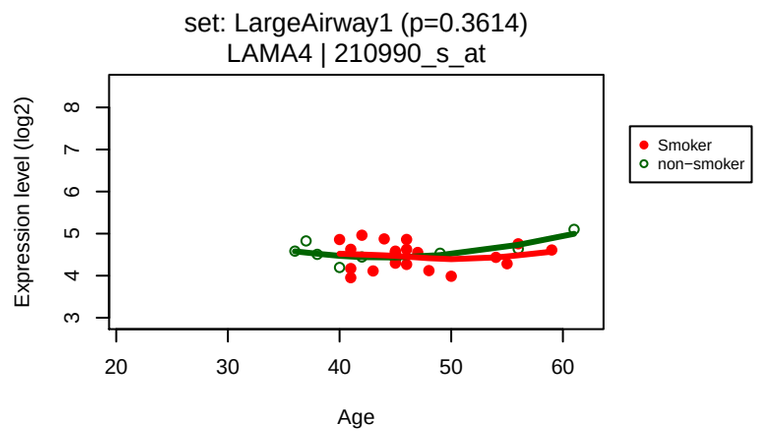

Gene expression of smokers and nonsmokers
gene: LAMA4 | 210990_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.304
p=0.634
p=0.1335
p=0.3614
p=0.9564
p=0.0329
p=0.1116
p=0.6605
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

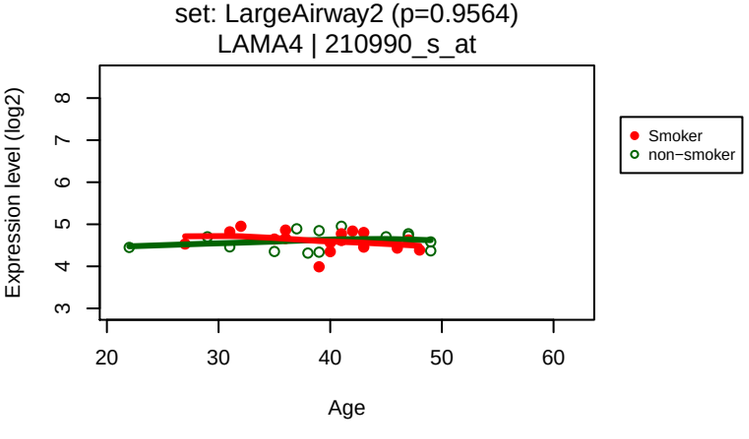

Gene expression ratio (smokers/nonsmokers)
gene: LAMA4 | 210990_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.304
p=0.634
p=0.1335
p=0.3614
p=0.9564
p=0.0329
p=0.1116
p=0.6605
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

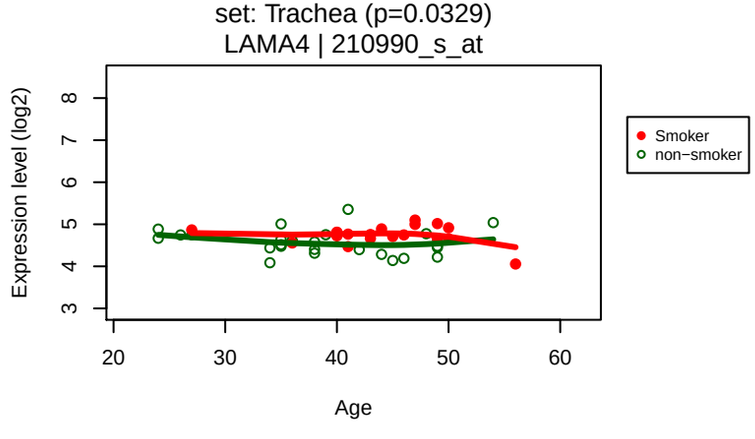

Gene expression ratio in Males
gene: LAMA4 | 210990_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7307
p=0.8525
p=0.016
p=0.9415
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LAMA4 | 210990_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.18
p=0.8875
p=0.3112
p=0.5361
SmallAirway2
LargeAirway2
Trachea
Macrophage2

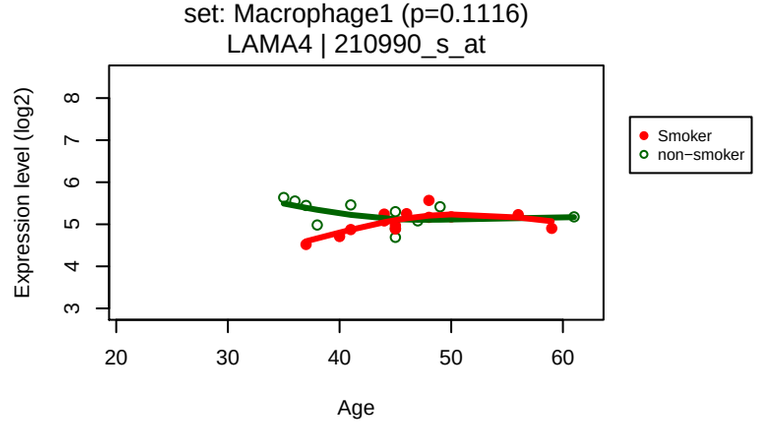

set: SmallAirway1 (p=0.304)
LAMA4 | 210990_s_at
Expression level (log2)
Age
Smoker
non−smoker

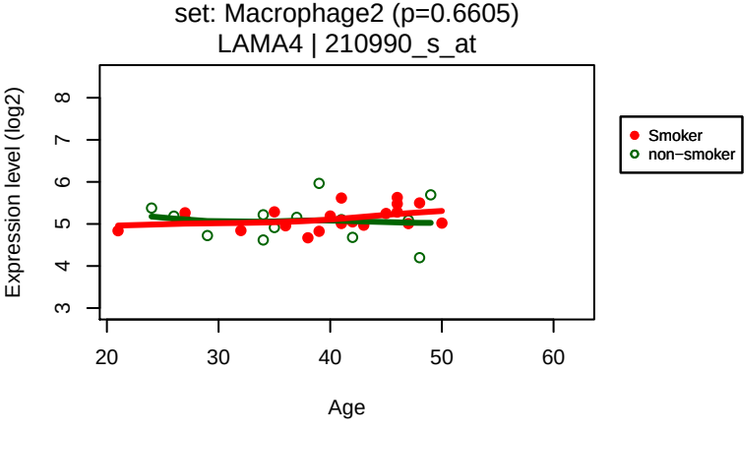

set: SmallAirway2 (p=0.634)
LAMA4 | 210990_s_at
Expression level (log2)
Age
Smoker
non−smoker

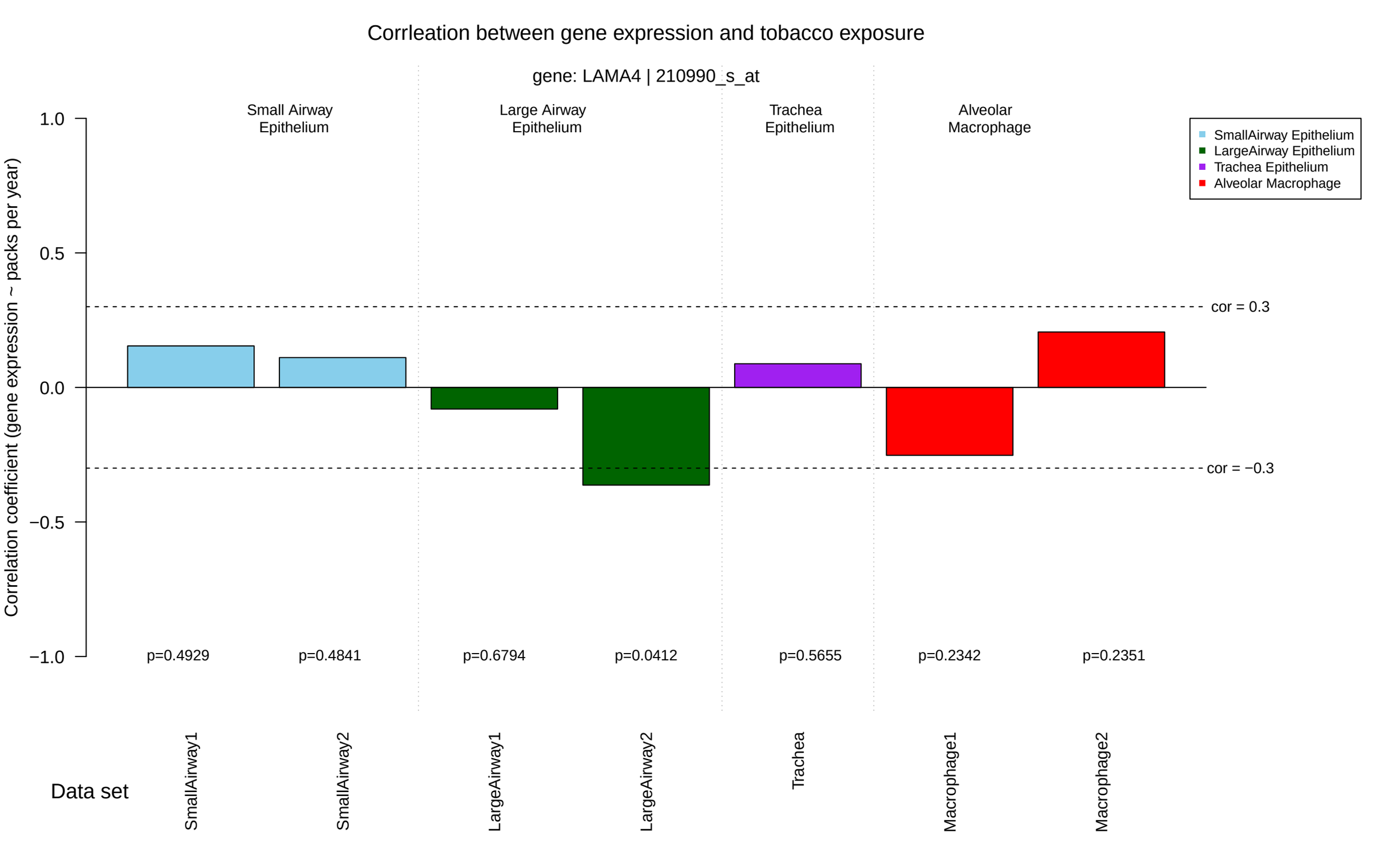

set: LargeAirway1 (p=0.3614)
LAMA4 | 210990_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9564)
LAMA4 | 210990_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0329)
LAMA4 | 210990_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1116)
LAMA4 | 210990_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6605)
LAMA4 | 210990_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LAMA4 | 210990_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4929
p=0.4841
p=0.6794
p=0.0412
p=0.5655
p=0.2342
p=0.2351
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2